Three-cloud propagation prototype: concurrency benchmark

NAUSICAA analysis

29 October 2026

Method

For each of 100 events from each 1 mm source sample (double beta and single electrons), the Julia prototype produced an ion cloud and electron clouds using the pure-Xe and Xe–15% He Petit diffusion coefficients. The event location was drawn once per source event; ion and electron distances sum to 150 cm. The benchmark includes Julia process startup, HDF5 source reads, diffusion and compressed HDF5 output. Each stream writes a separate file.

Results

Workers	Events/stream	Wall time (s)	Speedup	Output (MB)
1	100	39.03	1.00	64.2
2	100	19.98	1.95	64.2
4	100	13.59	2.87	64.2
6	100	7.63	5.12	64.2

The sum of individual peak resident-memory measurements for six processes is 4.90 GB. This sum is a conservative ceiling for simultaneous resident memory, not a measured concurrent peak.

Validation and limits

All 100 events in each source sample have the same sampled ion drift distance across their three clouds. Ion and electron drift distances sum to 150 cm. Every cloud conserves its sampled carriers across retained and outside-bin counts. Per-event arrays and counts match exactly for one, two, four and six workers. These measurements cover 100 events per stream and include output to the local filesystem; production-scale throughput and concurrent peak resident memory require a separate full-run measurement.